1552. Magnetic Force Between Two Balls (leet code Problem)

```
class Solution {
public:
    bool canPlace(vector<int>& a, int m, int d) {
        int cnt = 1, last = a[0];

        for (int i = 1; i < a.size(); i++) {
            if (a[i] - last >= d) {
                cnt++;
                last = a[i];
            }
            if (cnt == m) return true;
        }
        return false;
    }

    int maxDistance(vector<int>& a, int m) {
        sort(a.begin(), a.end());

        int l = 1, r = a.back() - a[0], ans = 0;

        while (l <= r) {
            int mid = l + (r - l) / 2;

            if (canPlace(a, m, mid)) {
                ans = mid;
                l = mid + 1;
            } else {
                r = mid - 1;
            }
        }
        return ans;
    }
};
```

Sum of Array | Practice | Geeksforc
Magnetic Force Between Two Balls

https://leetcode.com/problems/magnetic-force-between-two-balls/submissions/1906379795/

Import favorites | WhatsApp | U M S | ChatGPT | Introduction to Lar... | LPU LIVE - Authent... | LPU Live - Real-tim...

Problem List | Submit | 05:22:12 | Premium

Description | Accepted | Editorial | Solutions | Submissions

All Submissions

Accepted 98 / 98 testcases passed Time taken: 5 hrs 22 m 12 s

rahul_1252 submitted at Feb 03, 2026 12:29

Editorial | Solution

Runtime

44 ms | Beats 79.60%

Analyze Complexity

Memory

61.58 MB | Beats 94.01%

Code | C++

```
class Solution {
public:
```

Code

C++ | Auto

```cpp
class Solution {
public:
    bool canPlace(vector<int>& a, int m, int d) {
        int cnt = 1, last = a[0];

        for (int i = 1; i < a.size(); i++) {
            if (a[i] - last >= d) {
                cnt++;
                last = a[i];
            }
            if (cnt == m) return true;
        }
        return false;
    }

    int maxDistance(vector<int>& a, int m) {
        sort(a.begin(), a.end());

        int l = 1, r = a.back() - a[0], ans = 0;

        while (l <= r) {
            int mid = l + (r - l) / 2;

            if (canPlace(a, m, mid)) {
                ans = mid;
```

Saved | Ln 34, Col 1

Testcase | Test Result